

An infinite family of $3\vec{\omega}$ -critical tournaments and a negative answer to Question 5.9 of Aboulker, Aubian, Charbit and Lopes

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Machine-generated note

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Abstract

The clique number $\vec{\omega}(T)$ of a tournament T , introduced by Aboulker, Aubian, Charbit and Lopes, is the minimum over all linear orderings of $V(T)$ of the clique number of the backedge graph. They asked (Question 5.9) whether there is a function ℓ such that every tournament T with $\vec{\omega}(T) \geq k$ contains a subtournament on at most $\ell(k)$ vertices with clique number at least k . We answer this negatively, already for $k = 3$: for every $n \geq 3$ the circulant tournament $T_n = \text{Cay}(\mathbb{Z}_{2n+1}, \{1, \dots, n-1, n+1\})$ satisfies $\vec{\omega}(T_n) = 3$ while every proper subtournament has clique number at most 2. The family $(T_n)_{n \geq 3}$ also establishes the case $k = 3$ of their Conjecture 5.10 on the existence of infinitely many $k\vec{\omega}$ -critical tournaments. The same family and criticality result appear in Aubian and Coulomb [4, 6.1]. We retain this note as a self-contained account of a machine-generated proof of that result.

This note was produced by AI within the [Graph Theory AI](#) project: the argument was found by a language model and audited by an adversarial LLM referee, and was subsequently checked by human mathematicians. A machine-checked Rocq/MathComp formalization of this result is available as [question_5_9_disproved](#). Section 4 records the full provenance.

1 Introduction

Let D be a digraph and $\prec$ a linear ordering of $V(D)$. The *backedge graph* $D^\prec$ is the undirected graph on $V(D)$ in which uv is an edge whenever $u \prec v$ and vu is an arc of D (the arc goes “backwards” with respect to $\prec$). Aboulker, Aubian, Charbit and Lopes [1] define the *clique number* of D as

$$\vec{\omega}(D) = \min_{\prec} \omega(D^\prec),$$

the minimum over all orderings of the clique number of the backedge graph, and study the classes of tournaments whose dichromatic number is bounded by a function of $\vec{\omega}$. In the concluding section of [1] they ask whether a large clique number is always certified by a bounded-size subtournament:

Question 1.1 ([1, Question 5.9]). Is there a function ℓ such that, for every tournament T and every k , if $\vec{\omega}(T) \geq k$ then T has a subtournament A with $|A| \leq \ell(k)$ and $\vec{\omega}(A) \geq k$?

This strengthens their Conjecture 5.8, which allows the hypothesis $\vec{\omega}(T) \geq f(k)$ for a second function f ; that weaker statement was recently proved by Crew, Fan, Koerts, Moore and Spirkel [3, Corollary 7]. The authors of [1] also note that [Question 1.1](#) would be answered negatively by their Conjecture 5.10, which asserts that for every $k \geq 3$ there are infinitely many $k\vec{\omega}$ -critical tournaments, i.e. tournaments T with $\vec{\omega}(T) = k$ and $\vec{\omega}(T - v) = k - 1$ for every vertex v . (Since appending v at the end of an optimal ordering of $T - v$ raises the clique number of the backedge graph by at most one, $\vec{\omega}(T - v) \geq \vec{\omega}(T) - 1$ always holds, so “critical” is equivalent to $\vec{\omega}(T - v) < \vec{\omega}(T)$ for all v .) The

trivial critical tournaments are the single vertex for $k = 1$ and the directed triangle for $k = 2$. Chen and Wang [5, Theorem 1.1] have now proved Conjecture 5.10 for every $k \geq 3$.

We give an explicit infinite family of $3\text{-}\vec{\omega}$ -critical tournaments. Throughout, $\text{Cay}(\mathbb{Z}_q, S)$ for $S \subseteq \mathbb{Z}_q \setminus \{0\}$ with $S \cap (-S) = \emptyset$ and $S \cup (-S) = \mathbb{Z}_q \setminus \{0\}$ denotes the circulant tournament on $\mathbb{Z}_q$ with arcs $x \rightarrow y$ whenever $y - x \in S$.

Theorem 1.2 (Main theorem). *For every integer $n \geq 3$ let $q = 2n + 1$ and*

$$T_n = \text{Cay}(\mathbb{Z}_q, S_n), \quad S_n = \{1, 2, \dots, n-1, n+1\}.$$

Then T_n is a tournament on $2n + 1$ vertices with $\vec{\omega}(T_n) = 3$ and $\vec{\omega}(T_n - v) = 2$ for every vertex v . Consequently every proper subtournament of T_n has clique number at most 2, and T_n is $3\text{-}\vec{\omega}$ -critical.

Corollary 1.3. *The answer to Question 1.1 is negative: no function ℓ exists, and $k = 3$ already fails. Conjecture 5.10 of [1] holds for $k = 3$.*

Proof. If ℓ existed, choose $n \geq 3$ with $2n + 1 > \ell(3)$. By Theorem 1.2, $\vec{\omega}(T_n) = 3$ but every subtournament $A \subsetneq T_n$ has $\vec{\omega}(A) \leq 2$, so the only subtournament with clique number at least 3 is T_n itself, of order $2n + 1 > \ell(3)$. The second statement is immediate from Theorem 1.2. $\square$

Idea of proof. T_n is the standard rotational tournament $\text{Cay}(\mathbb{Z}_{2n+1}, \{1, \dots, n\})$ with the connection element n replaced by $n + 1$. Three short arguments do the work. (i) *Lower bound.* If every vertex of a tournament dominates a directed triangle, then in any ordering the last vertex v together with a backedge of that triangle forms a triangle of the backedge graph; in T_n every x dominates the directed triangle $\{x + 1, x + 2, x + n + 1\}$. (ii) *Upper bound.* In the natural ordering $0 \prec 1 \prec \dots \prec 2n$, the pair $i < j$ is a backedge iff $j - i \in \{n\} \cup \{n + 2, \dots, 2n\}$. Two consecutive gaps of a triangle would then each be at least n and sum to at most $2n$, forcing the triangle $\{0, n, 2n\}$; so this backedge graph has exactly one triangle and no K_4 . (iii) *Criticality.* Deleting the vertex 0 from the natural ordering deletes the unique triangle, so $\vec{\omega}(T_n - 0) \leq 2$; vertex-transitivity transfers this to every v , and monotonicity of $\vec{\omega}$ under taking subtournaments to every proper subtournament. The lower bound $\vec{\omega}(T_n - v) \geq 2$ holds because $T_n - v$ contains a directed triangle. [full proof]

2 Two general observations

Lemma 2.1. *If A is a subtournament of T then $\vec{\omega}(A) \leq \vec{\omega}(T)$. Moreover $\vec{\omega}(T) \geq 2$ whenever T contains a directed triangle.*

Idea of proof. Restricting an ordering of T to $V(A)$ gives the induced subgraph of the backedge graph on $V(A)$. A directed triangle is not transitive, so under any ordering one of its arcs is a backedge. [full proof]

Lemma 2.2 (Last-vertex lemma). *Let T be a tournament in which every vertex v has three out-neighbours inducing a directed triangle. Then $\vec{\omega}(T) \geq 3$.*

Idea of proof. Let v be the last vertex of an arbitrary ordering; all arcs $v \rightarrow a$ are backedges, and the directed triangle in $N^+(v)$ contributes one more backedge ab , so $\{v, a, b\}$ is a triangle of the backedge graph. [full proof]

3 Remarks

Remark 3.1 (Small cases and sharpness). $T_3 = \text{Cay}(\mathbb{Z}_7, \{1, 2, 4\})$ is the Paley tournament on 7 vertices. The referee's exhaustive search over all orderings confirms $\vec{\omega}(T_n) = 3$ and $\vec{\omega}(T_n - 0) = 2$ for $n = 3, \dots, 7$, and an independent SAT encoding confirms both for $n = 3, \dots, 30$, i.e. up to 61 vertices (Appendix B). The family shows that no bound $\ell(3)$ exists. This construction does not address $k \geq 4$, but those cases are covered by Chen and Wang [5, Theorem 1.1].

Remark 3.2 (Relation to complexity). Aubian [2] proved that deciding $\vec{\omega}(T) \leq k$ is NP-complete for every fixed $k \geq 3$. A positive answer to [Question 1.1](#) would place “ $\vec{\omega}(T) \leq 3$ ” in P (test all $O(|V|^{\ell(4)})$ subsets of size at most $\ell(4)$), so a negative answer was expected unless $P = NP$. [Theorem 1.2](#) gives an unconditional negative answer by an explicit family.

Remark 3.3 (Related work and status, 10 September 2026). We thank Samuel Coulomb for drawing our attention to recent work relevant to this note. The same circulant family constructed here, together with its $3\text{-}\vec{\omega}$ -criticality, appears in Aubian and Coulomb [4, 6.1]. Their result therefore also gives a negative answer to [Question 1.1](#). More generally, Chen and Wang [5, Theorem 1.1] establish the existence of infinitely many $k\text{-}\vec{\omega}$ -critical tournaments for every $k \geq 3$, settling Conjecture 5.10 of [1].

Accordingly, this note should be read as a self-contained account of a machine-generated proof of a result also established in the literature, rather than as a claim to a new resolution. The earlier novelty discussion and the statement that the cases $k \geq 4$ remain open are superseded by these references. The original referee report in [Appendix B](#) is retained unchanged as a historical record.

The provenance recorded in this note dates the original generation to 1 September 2026, before the 7 September submission of Aubian and Coulomb’s preprint. These dates document our workflow. They do not establish priority or independent discovery. Subsequent human review confirmed the proof and reported that the construction had also circulated privately before the machine run. We therefore make no claim of novelty or priority.

4 Provenance

The proof was generated, machine-refereed and rewritten without human intervention, in three stages. (1) The construction and proof were found by the language model gpt-5.6-sol in a single autonomous pass on 2026-09-01, as part of a large sweep over open problems catalogued from arXiv (catalog id 2310.04265__09; original writeup in `attacks/2310.04265__09/output.md`). (2) An independent adversarial referee agent (claude-fable-5) re-derived every step, checked the source paper, and verified the claims by exhaustive search for 7–15 vertices, by SAT solving up to 61 vertices and structurally up to 121 vertices; its verdict was CONFIRMED. Its report is reproduced verbatim in [Appendix B](#). (3) On 2026-09-09 the writeup was rewritten into the present note by claude-fable-5.1: the text was reorganised with full proofs in [Appendix A](#), the exact value $\vec{\omega}(T_n - v) = 2$ (needed for the word “critical” in the source paper’s sense, and noted as missing by the referee) was added, and context on related work was added. No other mathematical content was changed. (4) On 2026-09-10, the related-work discussion was updated following a report from Samuel Coulomb. The mathematical proof and the original referee report were left unchanged. (5) Independently of this note, the result was formalized in Rocq/MathComp in the companion repository [LLM4Rocq/graph-theory-rocq](#) as [question_5_9_disproved](#), checked against a source-verified formal statement with no added axioms. That check certifies the formal proposition, not its faithfulness to the source paper. (6) On 2026-09-18, the proof was checked by human mathematicians, who reported no mathematical error and noted that the construction was already known from earlier private discussions. The proof itself was left unchanged; only the review-status and prior-knowledge disclosures were updated.

A Full proofs

Lemma 2.1. *If A is a subtournament of T then $\vec{\omega}(A) \leq \vec{\omega}(T)$. Moreover $\vec{\omega}(T) \geq 2$ whenever T contains a directed triangle.*

Proof. Let $\prec$ be an ordering of $V(T)$ with $\omega(T^\prec) = \vec{\omega}(T)$ and let $\prec_A$ be its restriction to $V(A)$. For $u, v \in V(A)$, $uv \in E(A^{\prec_A})$ iff $u \prec v$ and $vu \in A(T)$ iff $uv \in E(T^\prec)$; so $A^{\prec_A}$ is the subgraph of $T^\prec$ induced by $V(A)$, and $\vec{\omega}(A) \leq \omega(A^{\prec_A}) \leq \omega(T^\prec) = \vec{\omega}(T)$.

If $a \rightarrow b \rightarrow c \rightarrow a$ is a directed triangle of T and $\prec$ is any ordering, the three vertices appear in some order, say $x \prec y \prec z$ with $\{x, y, z\} = \{a, b, c\}$; the arcs of the triangle form a directed cycle, so

they cannot all go forward (from the $\prec$ -smaller to the $\prec$ -larger vertex), and any backward arc gives an edge of $T^\prec$. Hence $\omega(T^\prec) \geq 2$ for every $\prec$. $\square$

Lemma 2.2 (Last-vertex lemma). *Let T be a tournament in which every vertex v has three out-neighbours inducing a directed triangle. Then $\vec{\omega}(T) \geq 3$.*

Proof. Let $\prec$ be an arbitrary ordering of $V(T)$ and let v be its last vertex. By hypothesis there are $a, b, c \in N^+(v)$ with $a \rightarrow b \rightarrow c \rightarrow a$. Since v is last, $a, b, c \prec v$, and since $v \rightarrow a, v \rightarrow b, v \rightarrow c$ are arcs, the pairs av, bv, cv are edges of $T^\prec$. As in the proof of Lemma 2.1, at least one arc of the directed triangle is a backedge, say (after renaming) $ab \in E(T^\prec)$. Then $\{a, b, v\}$ is a triangle in $T^\prec$, so $\omega(T^\prec) \geq 3$. This holds for every $\prec$, hence $\vec{\omega}(T) \geq 3$. $\square$

Theorem 1.2 (Main theorem). *For every integer $n \geq 3$ let $q = 2n + 1$ and*

$$T_n = \text{Cay}(\mathbb{Z}_q, S_n), \quad S_n = \{1, 2, \dots, n-1, n+1\}.$$

Then T_n is a tournament on $2n + 1$ vertices with $\vec{\omega}(T_n) = 3$ and $\vec{\omega}(T_n - v) = 2$ for every vertex v . Consequently every proper subtournament of T_n has clique number at most 2, and T_n is 3- $\vec{\omega}$ -critical.

Proof. Fix $n \geq 3, q = 2n + 1$, and work in $\mathbb{Z}_q$. Recall $S_n = \{1, \dots, n-1\} \cup \{n+1\}$.

T_n is a tournament. We have $-S_n = \{q-1, \dots, q-n+1\} \cup \{q-n-1\} = \{n+2, \dots, 2n\} \cup \{n\}$. Thus $S_n \cap (-S_n) = \emptyset$ and $S_n \cup (-S_n) = \{1, \dots, 2n\} = \mathbb{Z}_q \setminus \{0\}$, so for distinct x, y exactly one of $y-x \in S_n, x-y \in S_n$ holds: exactly one of the arcs $x \rightarrow y, y \rightarrow x$ is present. Translations $x \mapsto x+t$ are automorphisms of T_n , since arcs depend only on differences.

Lower bound $\vec{\omega}(T_n) \geq 3$. Let $x \in \mathbb{Z}_q$. The differences $1, 2, n+1$ lie in S_n (here $n \geq 3$ ensures $2 \leq n-1$), so $x \rightarrow x+1, x \rightarrow x+2, x \rightarrow x+n+1$. Moreover $(x+2) - (x+1) = 1 \in S_n$, $(x+n+1) - (x+2) = n-1 \in S_n$, and $(x+1) - (x+n+1) = -n \equiv n+1 \in S_n$; so $x+1 \rightarrow x+2 \rightarrow x+n+1 \rightarrow x+1$ is a directed triangle in $N^+(x)$. Lemma 2.2 gives $\vec{\omega}(T_n) \geq 3$.

Upper bound $\vec{\omega}(T_n) \leq 3$. Consider the natural ordering $0 \prec 1 \prec \dots \prec 2n$ and let $0 \leq i < j \leq 2n$, $d = j-i \in \{1, \dots, 2n\}$. The pair ij is an edge of the backedge graph iff $j \rightarrow i$, i.e. iff $i-j \equiv q-d \in S_n$, i.e. iff $q-d \in \{1, \dots, n-1\} \cup \{n+1\}$, i.e. iff

$$d \in D_n := \{n+2, \dots, 2n\} \cup \{n\}.$$

Suppose $i < j < k$ form a triangle of the backedge graph and put $a = j-i, b = k-j$. Then $a, b \in D_n$, so $a, b \geq n$, while $a+b = k-i \leq 2n$. Hence $a = b = n, k = i+2n \leq 2n$ forces $i = 0$, and $(i, j, k) = (0, n, 2n)$. Conversely the differences among $0, n, 2n$ are $n, n, 2n \in D_n$, so $\{0, n, 2n\}$ is a triangle. Therefore the backedge graph of the natural ordering has exactly one triangle and in particular no K_4 , so $\omega(T_n^\prec) = 3$ and $\vec{\omega}(T_n) \leq 3$. Together with the lower bound, $\vec{\omega}(T_n) = 3$.

$\vec{\omega}(T_n - v) = 2$ for every v . Restricting the natural ordering to $V(T_n) \setminus \{0\}$ gives, by the proof of Lemma 2.1, the backedge graph induced on $\{1, \dots, 2n\}$, which is triangle-free because the unique triangle $\{0, n, 2n\}$ contains 0. Hence $\vec{\omega}(T_n - 0) \leq 2$. For an arbitrary vertex v , the translation $x \mapsto x+v$ is an isomorphism from $T_n - 0$ to $T_n - v$, so $\vec{\omega}(T_n - v) \leq 2$. Finally $T_n - v$ contains a directed triangle: by the lower-bound paragraph, $\{v+1, v+2, v+n+1\}$ induces one and avoids v . By Lemma 2.1, $\vec{\omega}(T_n - v) \geq 2$.

Proper subtournaments. If $A \subsetneq T_n$, pick $v \in V(T_n) \setminus V(A)$; then A is a subtournament of $T_n - v$ and Lemma 2.1 gives $\vec{\omega}(A) \leq \vec{\omega}(T_n - v) = 2$. So T_n is 3- $\vec{\omega}$ -critical. $\square$

B Report of the LLM referee (verbatim)

The following is the unedited report of the adversarial referee agent (claude-fable-5, 2026-09-02) on the original writeup. Its step numbers refer to that writeup, not to this note. The scripts it mentions are in verification/scripts/2310.04265__09/. Treat it as machine output, not as an authority.

B.1 Interpretation

Question 5.9 of Aboulker–Aubian–Charbit–Lopes, “Clique number of tournaments” (arXiv:2310.04265), asks — quoting the paper verbatim, confirmed against the v2 full text — “Is there a function ℓ such that, for every tournament T , if $\omega(T) \geq k$, then T has a subtournament A such that $|A| \leq \ell(k)$ and $\omega(A) \geq k$?” Here $\omega(D) = \min$ over vertex orderings $<$ of the clique number of the backedge graph $D^{<}$, where $uv \in E(D^{<})$ iff $u < v$ and the arc $vu \in A(D)$. The writeup uses exactly this convention (Section 3 of the writeup: with $i < j$ in the ordering, ij is a backedge iff $j \rightarrow i$), so there is no convention mismatch.

The writeup’s formalization is the intended one. It refutes the question at $k = 3$ by exhibiting, for every $n \geq 3$, a tournament T_n on $2n+1$ vertices with $\omega(T_n) = 3$ whose every proper subtournament has $\omega \leq 2$; since no bounded value $\ell(3)$ can then cover all T_n , no function ℓ exists. This is exactly the refutation route the source-paper authors themselves anticipated: they state that Conjecture 5.10 (infinitely many k - ω -critical tournaments) would answer Question 5.9 negatively, and the writeup in effect proves the $k = 3$ case of Conjecture 5.10. A reasonable author of the source paper would consider Question 5.9 resolved (negatively) by this argument. No interpretation gaming.

One quantifier note: k is implicitly universally quantified in Question 5.9; a single failing k (here $k = 3$) suffices to kill ℓ , so the $k = 3$ refutation is logically adequate.

B.2 Step-by-step findings

step	label	note
1. Construction: $T_n = \text{Cay}(Z_{\{2n+1\}}, S_n)$, $S_n = \{1, \dots, n-1, n+1\}$ is a tournament	VALID	$-S_n = \{n, n+2, \dots, 2n\}$; $S_n \cap (-S_n) = \emptyset$ and $S_n \cup (-S_n) = Z_q \setminus \{0\}$. Re-derived by hand and verified by code for $n = 3..60$.
2. Translations are automorphisms of T_n	VALID	Standard for Cayley digraphs: arcs depend only on differences.
3. Lemma: if every vertex dominates a directed triangle then $\omega(T) \geq 3$	VALID	For any ordering, let v be its last vertex; all arcs $v \rightarrow a$ to out-neighbours are backedges. A directed triangle in $N^+(v)$ cannot be consistent with a linear order, so one of its arcs is a backedge, say giving edge ab ; then $\{v, a, b\}$ is a backedge-graph triangle. Airtight.
4. Every $x \in Z_q$ dominates the directed triangle $\{x+1, x+2, x+n+1\}$	VALID	Differences $1, 2, n+1 \in S_n$ (needs $n \geq 3$, which is assumed, so that $2 \leq n-1$); triangle arcs have differences $1, n-1$, and $-(n) \equiv n+1 \pmod{q}$, all in S_n . Verified by code for $n = 3..60$. Hence $\omega(T_n) \geq 3$.
5. Natural-order backedge graph is the circulant difference graph with $D_n = \{n\} \cup \{n+2, \dots, 2n\}$	VALID	ij ($i < j$, $d = j-i$) is a backedge iff $q-d \in S_n$, i.e. $d = n$ or $n+2 \leq d \leq 2n$. Verified edge-by-edge by code for $n = 3..60$.

step	label	note
6. That graph's unique triangle is $\{0, n, 2n\}$; no K_4 ; hence $\omega(\vec{T}_n) \leq 3$	VALID	If $i < j < k$ form a triangle, $a = j - i$ and $b = k - j$ lie in D_n so $a, b \geq n$, while $a + b = k - i \leq 2n$; thus $a = b = n$, forcing $i = 0$. Also $k - i = 2n \in D_n$ ($n \geq 2$). Code confirms exactly one triangle and no K_4 for $n = 3..60$. Combined with step 4: $\omega(\vec{T}_n) = 3$.
7. Restricting the natural order to $1 < \dots < 2n$ shows $\omega(\vec{T}_n - 0) \leq 2$	VALID	Restricting an ordering yields the induced backedge subgraph; the unique triangle used vertex 0, so the restriction is triangle-free. Code confirms triangle-freeness for $n = 3..60$. Follows from step 2.
8. Vertex-transitivity gives $\omega(\vec{T}_n - v) \leq 2$ for every v	VALID	
9. Monotonicity: $A \subseteq T$ implies $\omega(\vec{A}) \leq \omega(\vec{T})$; hence every proper subtournament has $\omega \leq 2$	VALID	Restricting the optimal ordering of T to $V(A)$ gives an induced subgraph of the optimal backedge graph. Any proper A avoids some v , so $\omega(\vec{A}) \leq \omega(\vec{T}_n - v) \leq 2$.
10. Disproof of Question 5.9 at $k = 3$	VALID	If ℓ existed, pick n with $2n+1 > \ell(3)$; the only subtournament of T_n with $\omega \geq 3$ is T_n itself, of order $2n+1 > \ell(3)$. Contradiction.

One trivial omission (does not affect the disproof): the writeup calls T_n “3- ω -critical”, which by the source paper’s definition requires $\omega(\vec{T}_n - v) = 2$ exactly; the writeup proves only ≤ 2 . The missing half is immediate — $T_n - v$ contains directed triangles for $n \geq 3$, so $\omega(\vec{T}_n - v) \geq 2$ — and my brute force confirms equality for $n = 3..7$. The disproof itself uses only the ≤ 2 direction.

B.3 Reference check

- **arXiv:2310.04265 (source paper), v2 of 2026-06-22.** Fetched the full HTML text. Question 5.9, Conjecture 5.8, Conjecture 5.10, and the backedge-graph definition of ω all state exactly what the prompt and writeup use, with the same convention (edge uv iff $u < v$ and $vu \in A(D)$). The paper knows only the trivial critical tournaments (single vertex for $k=1$, directed triangle for $k=2$) and gives no nontrivial critical families; v2 contains no note that Question 5.9 or Conjecture 5.10 has since been resolved.
- **arXiv:2602.09863 (Crew–Fan–Koerts–Moore–Spirk, Feb 2026).** Fetched. Its Corollary 7 is exactly the two-function statement ($\omega(\vec{T}) \geq f(k)$ implies a subtournament of size $\leq \ell(k)$ with $\omega \geq k$), i.e. Conjecture 5.8, which the paper explicitly says it confirms. It does not mention Question 5.9, Conjecture 5.10, or critical tournaments, and does not claim f can be the identity — so there is no conflict between its “bounded-size certificate” abstract phrasing and this disproof.
- **Priority sweep.** Searched for prior resolutions: arXiv:2401.07776 (Aubian, “Computing the clique number of tournaments”) proves deciding $\omega(\vec{T}) \leq k$ NP-complete for $k \geq 3$ and refutes Conjecture 4.1 (not 5.9/5.10) of the source paper; arXiv:2606.07748 (Aboulker et al., June 2026) and arXiv:2608.09710 (Aug 2026) cite the source paper but mention no resolution of Question 5.9

or Conjecture 5.10. I found no publication containing this construction or any unconditional negative answer to Question 5.9. Note that Aubian’s NP-completeness already implies a *conditional* negative answer (a positive answer to 5.9 would put “ $\omega \leq 3$ ” in P by testing all $O(n^{\ell(4)})$ subsets of size $\leq \ell(4)$, hence $P = NP$), so the community had strong reason to expect the negative answer; an unconditional disproof via an explicit 3-critical family nevertheless appears to be new.

B.4 Computational check

Scripts in `verification/scripts/2310.04265__09/` (`verify.py`, `sat_check.py`; kissat used as the SAT backend).

- **Structural checks, $n = 3..60$ (up to 121 vertices):** T_n is a tournament; every vertex dominates the claimed directed triangle $\{x+1, x+2, x+n+1\}$; the natural-order backedge graph equals the claimed circulant graph with differences $D_n = \{n\} \cup \{n+2, \dots, 2n\}$; its unique triangle is $\{0, n, 2n\}$ and it has no K_4 ; the restricted order on $T_n - 0$ gives a triangle-free backedge graph. All pass.
- **Exhaustive $\vec{\omega}$ over all orderings (pruned DFS, first vertex fixed to 0 by vertex-transitivity), $n = 3..7$ (7, 9, 11, 13, 15 vertices):** $\vec{\omega}(T_n) = 3$ and $\vec{\omega}(T_n - 0) = 2$ in every case. (For $n = 3$, T_3 is the Paley tournament on 7 vertices; $\vec{\omega} = 3$ confirmed.)
- **Independent SAT verification, $n = 3..30$ (7 to 61 vertices):** encoded “there exists an ordering whose backedge graph is triangle-free” as CNF (precedence variables + transitivity clauses + one clause per vertex triple); kissat reports UNSAT for every T_n (so $\vec{\omega}(T_n) \geq 3$ for all orderings, matching the Lemma) and SAT for every $T_n - 0$ (a triangle-free ordering exists, so $\vec{\omega}(T_n - 0) \leq 2$). All 28 instances complete in ~ 8 s total. No discrepancy of any kind was found.

B.5 Caveats

1. **Criticality half-stated.** “3- $\vec{\omega}$ -critical” formally needs $\vec{\omega}(T_n - v) = 2$, not just ≤ 2 ; the ≥ 2 half ($T_n - v$ contains a directed cycle) is trivial and computationally confirmed, and the disproof only uses ≤ 2 . Cosmetic.
2. **Range $n \geq 3$ is necessary** (the argument uses $2 \in S_n$, i.e. $2 \leq n-1$); the writeup states this hypothesis, so no gap — it merely means the family starts at 7 vertices.
3. **Convention dependence.** The proof is tied to the source paper’s backedge convention (edge iff the arc goes right-to-left in the ordering); the writeup matches it exactly. Reversing all arcs / orders gives the same parameter, so nothing hinges on this.
4. **Scope vs. claim.** The verdict line says Question 5.9 is disproved; that is exactly what is shown. The construction additionally establishes the $k = 3$ case of Conjecture 5.10 but not the full conjecture ($k \geq 4$ remains open); the writeup does not overclaim this.
5. **Novelty.** The writeup itself flags that literature priority was not checked. My sweep (through papers up to Aug 2026 by the same research community) found no prior unconditional disproof or any appearance of this family; the closest prior art is the $P \neq NP$ -conditional negative answer implicit in Aubian’s NP-completeness result (arXiv:2401.07776). I cannot fully exclude an unpublished or missed source, but ALREADY_KNOWN is not supported by anything I found.

B.6 Referee summary

I set out to break this proof and could not. The construction is a one-line circulant tweak of the standard rotational tournament (replace difference n by $n+1$ in $S = \{1, \dots, n\}$), and every claim about it survives full re-derivation and independent computation: the last-vertex lemma is airtight, the natural-order backedge graph analysis is a correct two-line additive argument, and criticality follows cleanly from vertex-transitivity plus monotonicity. Exhaustive search over all orderings ($n = 3..7$) and an independent SAT encoding ($n = 3..30$, up to 61 vertices) confirm $\vec{\omega}(T_n) = 3$ and $\vec{\omega}(T_n - 0) = 2$ with zero discrepancies; structural claims were machine-checked to $n = 60$. The interpretation matches the

source paper verbatim, all citations check out, and the $k = 3$ refutation is logically sufficient against the universally quantified question. The disproof is correct.

References

- [1] P. Aboulker, G. Aubian, P. Charbit and R. Lopes, Clique number of tournaments, arXiv:2310.04265 (2023; v2 June 2026).
- [2] G. Aubian, Computing the clique number of tournaments, arXiv:2401.07776 (2024).
- [3] L. Crew, Y. Fan, H. Koerts, B. Moore and S. Spirkl, arXiv:2602.09863 (2026), Corollary 7.
- [4] G. Aubian and S. Coulomb, Clique Number of Tournaments II, [arXiv:2609.07481](#) (2026), result 6.1. Submitted 7 September 2026.
- [5] G. Chen and S. Wang, Cayley Tournaments Simultaneously Critical for the Clique and Dichromatic Numbers, [arXiv:2609.08658](#) (2026), Theorem 1.1.